Section A: Pure Mathematics [40 marks]

1 Find the following integrals

(a) $\int (5-x)(2x+1)\ \mathrm{d}x$, [2]

(b) $\int \frac{xe^{3x}-2e^{3x}+1}{x-2}\ \mathrm{d}x$. [2]

2 The function f is defined by $\mathrm{f}: x \mapsto x^2-3x+2,\ x \in \mathbb{R},\quad x>0$.

(i) Show that the inverse function, f^{-1}, does not exist. [2]

(ii) Find the largest possible domain of f, in the form $[k,\infty)$, such that f^{-1} exists. Express f^{-1} in similar form and state its range. [5]

3 **(i)** Sketch the graph of $y=\frac{ax-1}{x-1}$ where $a>0$. Write down the coordinates of the axial intercepts and the equations of the asymptotes, leaving your answers in terms of *a*. [3]

(ii) Express $y=\frac{ax-1}{x-1}$ in the form $y=\frac{m\,(x-1)+n}{x-1}$, where *m* and *n* are expressed in terms of *a*. [2]

(iii) Given that the area bounded by the curve $y=\frac{ax-1}{x-1}$, the *x*-axis, the lines $x=-2$ and $x=0$ is $4-\ln 3$, show that $a=2$. [4]

4 **(a)** Sketch on a single diagram, the graphs of $2y-1=x$ and $y=\left|x^2-1\right|$.

Hence or otherwise, solve the inequality $2\left|x^2-1\right|>x+1$. [6]

(b) Find the range of values of *x* for which $3x-x^2 \le -4$.

Hence solve the inequality $3|x|-x^2 \le -4$. [4]

5 **(a)** Find the equation of the normal to the curve $y=\ln x^2$ at $x=\mathrm{e}$. [3]

(b) The gradient of the curve $y=ax+\frac{b}{x^2}$ at the point (1, 10) is −14.

(i) Find the values of *a* and *b*. [4]

(ii) Using a non-calculator method, find the coordinates of the stationary point on the curve and determine the nature of the stationary point. [3]

Section B: Statistics [60 marks]

6 Analysis of the results of a certain group of students who had taken examinations in Mathematics and Chemistry produced the following information: 85% of the students passed in Mathematics, 80% passed in Chemistry and 25% failed in at least one of these subjects.

(i) Show that 75% of the students passed both Mathematics and Chemistry. [1]

(ii) Find the percentage of students who passed in exactly one of the two subjects. [2]

(iii) Of the students who passed in Mathematics, find the percentage who also passed in Chemistry. [2]

(iv) Determine whether the results of the Mathematics examination and Chemistry examination for this group of students are independent. [2]

7 The mass of children in a village may be taken to have a normal distribution with mean 53.5 kg. Given that 90% of the children have mass greater than 40 kg, find the standard deviation of the mass of children. [3]

Find the smallest number of children required if the probability that at least one of them has mass greater than 53.5 kg exceeds 0.99. [4]

8 **(a)** A manufacturer tests the quality of the containers he produces by taking a random sample of the containers.

(i) Explain clearly how the manufacturer would choose a systematic sample of 50 containers from a population of 250 containers. [2]

(ii) State one advantage and one disadvantage of systematic sampling method used in this context. [2]

(iii) If the containers are made in two different sizes, 'Large' and 'Small', in the ratio 3:2, how would the manufacturer select a stratified sample if he chooses to test 50 containers. [2]

(b) The random variable X has a normal distribution with mean 4 and standard deviation 10. If $\overline{X}$ is the mean of a sample of 12 independent observations of X, find the value of a such that $\mathrm{P}(\overline{X} < a) = 0.75$. [2]

9 Bottles of a particular brand of liquid hand soap are said to contain 400 ml. A random sample of 100 bottles is taken and the volumes of liquid in the bottles are measured. The volumes, x ml, are summarised by $\sum(x-400) = -220$ and $\sum(x-400)^2 = 7416$.

(i) Find unbiased estimates of the population mean and variance. [3]

(ii) Assuming a normal distribution, test at the 5% significance level whether the population mean volume is less than 400 ml. [4]

(iii) State, giving a reason, whether it is necessary to assume a normal distribution for the test to be valid. [1]

10 A company produces electrical components and the probability of it being defective is p. It is found that the mean number of defective electrical components in a batch of n electrical components is 2.

(i) Given that the variance of the number of electrical components that are defective in a batch of n electrical components is 1.8, find n and p. [3]

(ii) Show that the probability that a batch of n electrical components containing no defective electrical components is 0.122. [1]

(iii) 50 such batches of n electrical components are checked for defective components. Using a suitable approximation, find the probability that more than 10 batches out of the 50 contain no defective electrical components. [5]

11 Cucumbers are stored in brine before being processed into pickles. Data are calculated on x, the percentage of sodium chloride in the brine, and y, a measure of the firmness of the pickles produced. The data are shown below:

x	6.0	6.5	7.0	7.5	8.0	8.5	9.0	9.5
y	15.3	15.8	16.1	16.7	k	17.8	18.2	18.3

(i) Given that the regression line of y on x is $y = 9.80 + 0.921x$, verify that the value of k is 17.3. [3]

(ii) Give a sketch of the scatter diagram for the data, as shown on your calculator. State the value of the product moment correlation coefficient. [3]

(iii) Estimate a measure of the firmness of the pickles when the percentage of sodium chloride in the brine is 10. Comment on the reliability of the estimate obtained. [3]

12 A public examination in music consists of two papers, one theoretical and one practical. From past records the marks, X, for theoretical paper has mean 52 and standard deviation 10. The marks, Y, for practical paper has mean 48 and standard deviation 12. Assume that the distribution of marks follows a normal distribution and performances in both papers are independent.

Find the probability that a candidate chosen at random

(a) scores higher in the practical paper than in the theoretical paper, [3]

(b) scores more than 48 marks in both papers, [2]

(c) scores more than a total of 96 marks in both papers. [3]

Explain why the answer to part (c) is greater than the answer to part (b). [1]

It is later decided that a candidate needs to score an average of at least 45 marks in order to pass the music examination. Find the percentage of students who will pass the music examination if the average mark is defined as $\frac{1}{2}\left(X+Y\right)$. [3]

End of Paper